General Bite #2: Antenna Structures, Beacons, and Good Practice

Antenna height limits, FAA notification, beacon stations, one-way transmissions, state and local regulation, and the 'good engineering' rule.

Group G1B · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Antenna Structure Height Limits

What is the maximum height above ground for an antenna structure not near a public-use airport, without requiring FAA notification and FCC registration? 200 feet (G1B01).
Above 200 ft requires FAA notification (FAA Form 7460-1) and FCC antenna structure registration. Near airports the rules are tighter (the FAA glide slopes kick in earlier).

State and Local Regulation (PRB-1)

Under what conditions are state and local governments permitted to regulate amateur radio antenna structures? Amateur Service communications must be reasonably accommodated, and regulations must constitute the minimum practical to accommodate a legitimate purpose of the state or local entity. (G1B06)
This is the **PRB-1** federal preemption: states and localities can regulate, but they cannot effectively ban amateur antennas. Reasonable accommodation is required.

Beacon Stations

A beacon is an automatic, one-way transmission used by other amateurs to gauge propagation conditions.

Question	Answer	Q
Purpose of a beacon station	**Observation of propagation and reception** (G1B03).	G1B03
Beacon-station rules	**No more than one beacon station may transmit in the same band from the same station location** (G1B02).	G1B02
HF frequencies for auto-controlled beacons	**28.20 MHz to 28.30 MHz** (G1B09). The dedicated 10 m beacon segment.	G1B09
Beacon power limit	**100 watts PEP output** (G1B10).	G1B10

One-Way Transmissions And Permitted Content

Question	Answer	Q
Which one-way transmissions are permitted?	**Transmissions to assist with learning the International Morse code** (G1B05). Code-practice transmissions are explicitly allowed.	G1B05
Which transmission is permitted for all amateur stations?	**Occasional retransmission of weather and propagation forecast information from US government stations** (G1B04).	G1B04

International Operating

When may you communicate with amateur stations outside the FCC-administered areas? When the contact is with amateurs in any country except those whose administrations have notified the ITU that they object to such communications. (G1B08)
Default is allowed. The exception list is whoever specifically objected through the ITU.

Abbreviations And Procedural Signals

What are the restrictions on abbreviations or Q-signals? They may be used if they do not obscure the meaning of a message. (G1B07) Q-signals, CW shorthand, etc. are fine – just do not use them as a substitute code to hide content.

Who Decides “Good Practice”?

Who or what determines “good engineering and good amateur practice” for things Part 97 does not explicitly cover? The FCC. (G1B11) The FCC has final say.

Mental Cheat Sheet

- Antenna height limit (no FAA needed, not near airport): **200 ft**.
- PRB-1: state/local rules must **reasonably accommodate** amateur radio.
- Beacon purpose: **propagation observation**.
- Beacon power: **100 W PEP**.

Common Traps

- 200 ft is the trigger height – not 50, 100, or 250.
- PRB-1 does NOT say “no limits at all”; it requires **reasonable accommodation**.
- Beacon power is **100 W** PEP, not 10, 20, or 200.
- Beacon HF segment is **28.20-28.30**, not 21.08-21.09.

- Beacon HF segment: **28.20 to 28.30 MHz**.
- One station = at most **one beacon per band** at that location.
- Code-practice transmissions are **permitted** one-way ops.
- Re-transmitting NOAA/govt weather forecasts is **permitted**.
- DX contacts default **allowed** unless a country has objected via ITU.
- Q-signals and abbreviations OK as long as they don't **obscure** meaning.
- The **FCC** decides "good engineering and good amateur practice."

- The "good practice" arbiter is the **FCC**, not the control operator, IEEE, or ITU.
- Abbreviations are **allowed** with the no-obscuring rule; the trap answer says they are not permitted at all.
- International contact default is **allowed**; only specific objecting countries are off-limits.

Takeaway: The 200 ft / FAA / FCC registration rule, the PRB-1 reasonable-accommodation principle, and the beacon-station basics (purpose, power limit, 28.2-28.3 MHz HF segment) carry most of G1B. The FCC is the final arbiter for anything not explicitly written in Part 97.

Bite 2 Question Bank (G1B)

G1B01. What is the maximum height above ground for an antenna structure not near a public use airport without requiring notification to the FAA and registration with the FCC?
A. 50 feet.
B. 100 feet.
C. 200 feet.
D. 250 feet.

G1B02. With which of the following conditions must beacon stations comply?
A. No more than one beacon station may transmit in the same band from the same station location.
B. The frequency must be coordinated with the National Beacon Organization.
C. The frequency must be posted on the internet or published in a national periodical.
D. All these choices are correct.

G1B03. Which of the following is a purpose of a beacon station as identified in the FCC rules?
A. Observation of propagation and reception.
B. Automatic identification of repeaters.
C. Transmission of bulletins of general interest to amateur radio licensees.
D. All these choices are correct.

G1B04. Which of the following transmissions is permitted for all amateur stations?
A. Unidentified transmissions of less than 10 seconds duration for test purposes only.
B. Automatic retransmission of other amateur signals by any amateur station.
C. Occasional retransmission of weather and propagation forecast information from US government stations.
D. Encrypted messages, if not intended to facilitate a criminal act.

G1B05. Which of the following one-way transmissions are permitted?
A. Unidentified test transmissions of less than 10 seconds in duration.
B. Transmissions to assist with learning the International Morse code.
C. Regular transmissions offering equipment for sale, if intended for amateur radio use.
D. All these choices are correct.

G1B06. Under what conditions are state and local governments permitted to regulate amateur radio antenna structures?
A. Under no circumstances, FCC rules take priority.
B. At any time and to any extent necessary to accomplish a legitimate purpose of the state or local entity, provided that proper filings are made with the FCC.
C. Only when such structures exceed 50 feet in height and are clearly visible 1,000 feet from the structure.
D. Amateur Service communications must be reasonably accommodated, and regulations must constitute the minimum practical to accommodate a legitimate purpose of the state or local entity.

G1B07. What are the restrictions on the use of abbreviations or procedural signals in the amateur service?
A. Only "Q" signals are permitted.
B. They may be used if they do not obscure the meaning of a message.
C. They are not permitted.
D. They are limited to those expressly listed in Part 97 of the FCC rules.

G1B08. When is it permissible to communicate with amateur stations in countries outside the areas administered by the Federal Communications Commission?
A. Only when the foreign country has a formal third-party agreement filed with the FCC.
B. When the contact is with amateurs in any country except those whose administrations have notified the ITU that they object to such communications.
C. Only when the contact is with amateurs licensed by a country which is a member of the United Nations, or by a territory possessed by such a country.
D. Only when the contact is with amateurs licensed by a country which is a member of the International Amateur Radio Union, or by a territory possessed by such a country.

G1B09. On what HF frequencies are automatically controlled beacons permitted?
A. On any frequency if power is less than 1 watt.
B. On any frequency if transmissions are in Morse code.
C. 21.08 MHz to 21.09 MHz.
D. 28.20 MHz to 28.30 MHz.

G1B10. What is the power limit for beacon stations?
A. 10 watts PEP output.
B. 20 watts PEP output.
C. 100 watts PEP output.
D. 200 watts PEP output.

G1B11. Who or what determines "good engineering and good amateur practice," as applied to the operation of an amateur station in all respects not covered by the Part 97 rules?
A. The FCC.
B. The control operator.
C. The IEEE.
D. The ITU.

Answer Key

G1B01: C	G1B02: A	G1B03: A	G1B04: C
G1B05: B	G1B06: D	G1B07: B	G1B08: B
G1B09: D	G1B10: C	G1B11: A	